

Welfare and Spending Effects of Consumption Stimulus Policies

Christopher D. Carroll (JHU) Edmund Crawley (FED) William Du (JHU)
Ivan Frankovic (BBK) Håkon Tretvoll (SSB)

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- ▶ **This paper:** Aim to evaluate three consumption stimulus policies in a HA model consistent with data on liquid wealth and *intertemporal* MPCs

Related literature

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- ▶ **High MPCs and impatience:** Parker (2017)

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- ▶ Robustness Exercise: HANK model

Quantitative Micro Realism

- ▶ Idiosyncratic income process: Friedman/Muth (transitory and permanent shocks)

$\mathbf{p}$ — ‘permanent income’

ξ — ‘transitory income shock’

ψ — ‘permanent income shock’

$$\mathbf{p}_{t+1} = \mathcal{G}^e \mathbf{p}_t \psi_{t+1}$$

$$y_{t+1} = \mathbf{p}_{t+1} \xi_{t+1}$$

- ▶ $\mathcal{G}^e$: education-specific income growth
- ▶ Evidence for permanent shocks: See Crawley, Holm, and Tretvoll (2024)

Preferences, Beliefs, and Wealth

Infinite horizon model: target wealth depends on 'Growth Impatience' condition:

$$\underbrace{\left(\frac{(R \beta^{e,i})^{1/\gamma}}{\mathcal{G}^e \mathbb{E}[\psi^{-1}]} \right)}_{\text{'Growth Patience Factor'}} < 1 \quad (1)$$

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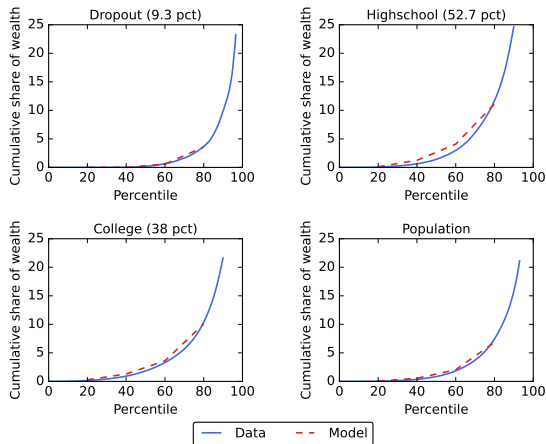
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Consistency With Micro Evidence (1)

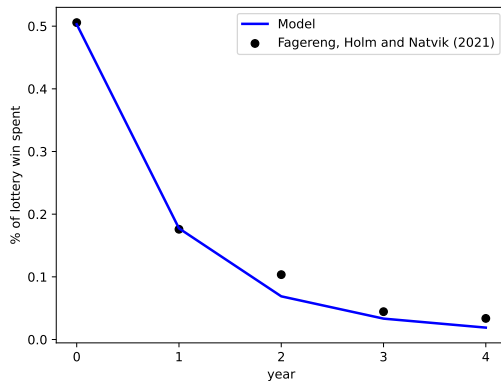
Liquid Wealth from
Survey of Consumer
Finances (SCF)



- ▶ Education groups: $e \in \{\text{'Dropout'}, \text{'Highschool' and 'College'}\}$
- ▶ Each group has distribution of discount factors $\beta_{e,i}$

Consistency With Micro Evidence (2)

Intertemporal MPC from Fagereng, Holm, Natvik (2021)



Modeling device: 'Splurge' in consumption

Splurge consumption

- ▶ Exogenous fraction of income directly consumed
- ▶ Model consistent with spending patterns over time after a transitory income shock
- ▶ Evidence: High liquid wealth hh also have high MPCs
 - ▶ Kueng (2018); Crawley and Kuchler (2023); Graham and McDowall (2024)
- ▶ Possible microfoundations:
 - ▶ Spending on durables (Browning and Crossley, 2009; Laibson et al., 2022)
 - ▶ A form of present bias (Indarte et al., 2024, Maxted et al., 2024)
- ▶ Robustness: Model w/o splurge consumption

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 - ▶ Welfare (only recession-related welfare impact)

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 - ▶ Very similar pattern for cumulative multipliers

Model

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- ▶ Remaining consumption $\mathbf{c}_{opt,i,t}$ is chosen to maximize utility

$$\sum_{t=0}^{\infty} \beta_{e,i}^t (1 - D)^t \mathbb{E}_0 u(\mathbf{c}_{opt,i,t}). \quad (2)$$

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- ▶ Budget constraint, given existing market resources $\mathbf{m}_{i,t}$ and income state, and a no-borrowing constraint:

$$\mathbf{m}_{i,t+1} = R \underbrace{(\mathbf{m}_{i,t} - \mathbf{c}_{sp,i,t} - \mathbf{c}_{opt,i,t})}_{\geq 0 \text{ (no-borrowing constraint)}} + \mathbf{y}_{i,t+1} \quad (3)$$

(R : exogenous gross interest rate)

Income process

- Income subject to transitory, unempl. and permanent shocks

$$\mathbf{y}_{i,t} = \begin{cases} \xi_{i,t}\mathbf{p}_{i,t}, & \text{if employed} \\ 0.7\mathbf{p}_{i,t}, & \text{if unemployed for } \leq 2q \\ 0.5\mathbf{p}_{i,t}, & \text{if unemployed } \geq 2q \end{cases} \quad (4)$$

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- ▶ Model is a simplified model of households (no heterogeneity in hh size)
- ▶ Replacement rates reflect some degree of hh insurance (Rothstein and Valetta, 2017)

Employment status and recessions

- ▶ Employment status is subject to a Markov process
 - ▶ Employed consumer: continue being employed or become unemployed
 - ▶ Unemployed consumers: receives benefits for two quarters
- ▶ Bureau of Labor Statistics: Report unemployment rates by education group
- ▶ Recession is given by an MIT shock
 - ▶ Unemployment rate doubles in each education group
 - ▶ Expected length of unemployment increases from 2 to 4q
 - ▶ End of recession occurs as a Bernoulli process calibrated for an avg. rec. length of 6q

Aggregate demand effects

(as in Krueger, Mitman and Perri, 2016)

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- ▶ Idiosyncratic income in the extension model is then given by

$$\mathbf{y}_{AD,i,t} = AD(C_t)\mathbf{y}_{i,t}. \quad (6)$$

Parametrization

Parametrization — Strategy

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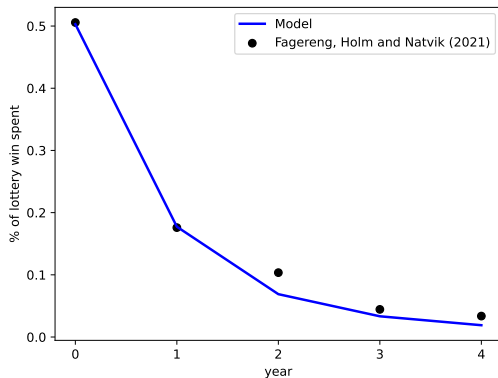
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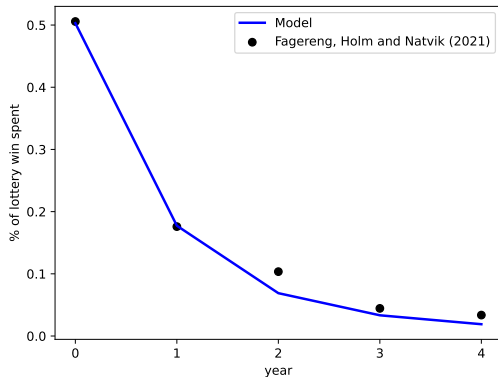
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 - ▶ β_e : center of discount factor distribution
 - ▶ ∇_e : spread of discount factor distribution
 - ▶ Uniform distribution, approximated with 7 different types

Step 1: iMPC from FHN (2021)



- Estimated splurge factor: $\varsigma = 0.31$; MPC across wealth distribution and K/Y untargeted but close to targets

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- ▶ Estimated splurge factor: $\varsigma = 0.31$; MPC across wealth distribution and K/Y untargeted but close to targets
- ▶ Zero splurge ($\varsigma = 0$): cannot match iMPC, wealth-dep. MPCs and K/Y-ratio at the same time

Parameters — same for all types Policy parameters

Parameters that apply to all types			
Parameter	Notation	Value	
Risk aversion	γ	2.0	
Splurge	ς	0.249	
Survival probability, quarterly	$1 - D$	0.994	
Risk free interest rate, quarterly (gross)	R	1.01	
Standard deviation of transitory shock	σ_{ξ}	0.346	
Standard deviation of permanent shock	σ_{ψ}	0.0548	
Unemployment benefits replacement rate (share of PI)	ρ_b	0	7
Unemployment income w/o benefits (share of PI)	ρ_{nb}	0	5
Avg. duration of unemp. benefits in normal times (quarters)		2	
Avg. duration of unemp. spell in normal times (quarters)		1.5	
Probability of leaving unemployment	π_{ue}	0.667	
Consumption elasticity of aggregate demand effect	κ	0.3	

Step 2b: Parameters — by education group

Parameters calibrated for each education group			
Dropout	Highschool	College	
Percent of population	9.3	52.7	38.0
Avg. quarterly PI of “newborn” agent (\$1000)	6.2	11.1	14.5
Std. dev. of log(PI) of “newborn” agent	0.32	0.42	0.53
Avg. quarterly gross growth rate of PI (Γ_e)	1.0036	1.0045	1.0049
Unemployment rate in normal times (percent)	8.5	4.4	2.7
Probability of entering unemployment (π_{eu}^e , percent)	6.2	3.1	1.8

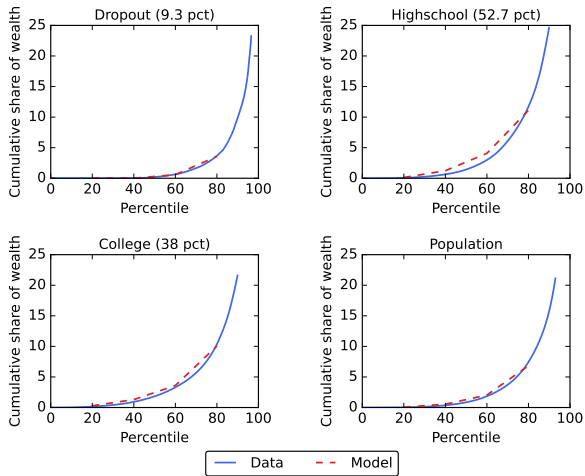
Step 3: Estimation of discount factors

	Dropout	Highschool	College
(β_e, ∇_e)	(0.719, 0.318)	(0.925, 0.077)	(0.983, 0.014)
(Min, max) in approximation	(0.447, 0.991)	(0.859, 0.990)	(0.971, 0.995)

Estimation targets	Dropout	Highschool	College
Median LW/ quarterly PI (data, percent)	4.64	30.2	112.8
Median LW/ quarterly PI (model, percent)	4.64	30.2	112.8

Non-targeted moments	Dropout	Highschool	College	Population
Percent of total wealth (data)	0.8	17.9	81.2	100
Percent of total wealth (model)	1.2	16.8	82.0	100
Avg. annual MPC (model, incl. splurge)	0.78	0.61	0.38	0.54

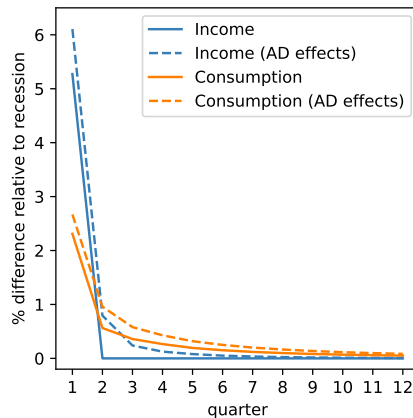
Step 3: Visualization of match with SCF



Results

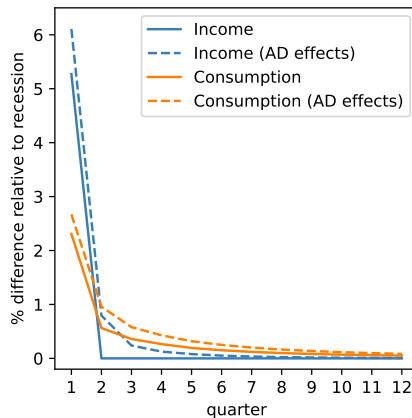
IRFs for stimulus check

- Simulate check policy in recessions lasting from 1 to 20 q



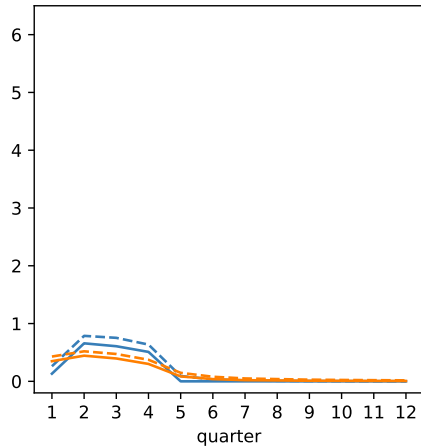
IRFs for stimulus check

- ▶ Simulate check policy in recessions lasting from 1 to 20 q
- ▶ Construct probability-weighted sum across rec. lengths

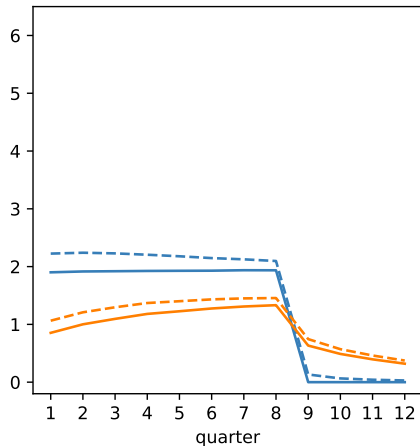


IRfs for extension of unemployment benefits / payroll tax cut

Extension of UI benefits:



Payroll tax cut:

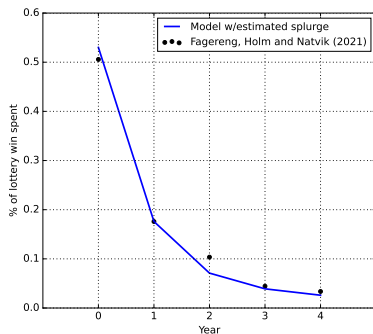


Untargeted moments (1)

Non-targeted moments by wealth quartile

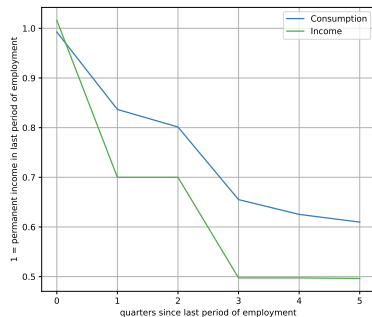
	WQ 4	WQ 3	WQ 2	WQ 1
Percent of liquid wealth (data)	0.14	1.60	8.51	89.76
Percent of liquid wealth (model, baseline)	0.12	0.98	3.85	95.0
Percent of liquid wealth (model, Splurge=0)	0.10	1.07	4.24	94.60
Avg. lottery-win-year MPC (model, incl. splurge)	0.74	0.61	0.48	0.32
Avg. lottery-win-year MPC (model, splurge=0)	0.69	0.53	0.36	0.14

Untargeted moments (2)



Share of lottery win spent

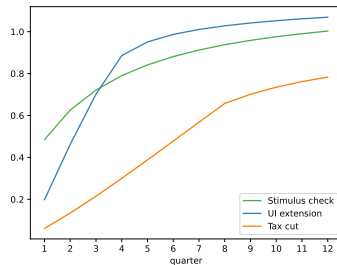
- Ganong and Noel (2019): UI expiry $\Rightarrow$ drop of 12 percent (month)
- Our model $\Rightarrow$ drop of 18 percent (quarter)



Spending upon expiry of UI benefits

Multipliers

$$M_t^P = \frac{\text{NPV of induced consumption up to } t}{\text{NPV of the cost of the policy}}$$



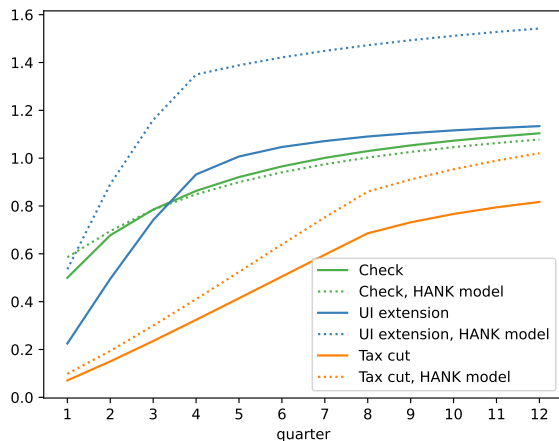
Spending results

	Stimulus check	UI extension	Tax cut
10y-horizon Multiplier (no AD effect)	0.88	0.91	0.85
10y-horizon Multiplier (AD effect)	1.23	1.21	0.98
Share of policy expenditure during recession	100.0%	79.6%	57.8%

Robustness: Multipliers in a HANK and SAM model — Setup

- ▶ Evaluate the policies in a relatively standard HANK and SAM model (Du, 2024)
- ▶ New Keynesian: Monopolistic competition + sticky prices
- ▶ Search and matching: Random search, labor market tightness affects job finding and vacancy filling probabilities
- ▶ Government policy: Monetary and fiscal rules
- ▶ Fiscal multipliers through an intertemporal Keynesian cross mechanism
However: No state dependence
- ▶ Solution method $\Rightarrow$ cannot evaluate effects starting in a deep recessionary state
This also implies that we cannot use our welfare measure

Robustness: Multipliers in a HANK and SAM model — Results



HA w/AD effects + HANK and SAM

Welfare measure

- Aim: Welfare measure does not reflect benefits of redistribution in “normal” times

$$\mathcal{W}(\text{policy}, Rec, AD) = \frac{1}{\mathcal{N}} \sum_{i=1}^N \sum_{t=0}^{\infty} \frac{1}{R^t} \frac{u(\mathbf{c}_{it, \text{policy}, Rec, AD}) - u(\mathbf{c}_{it, \text{none}, Rec, AD})}{u'(\mathbf{c}_{it, \text{normal}})}$$

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where $\mathcal{N} = NPV(\text{policy}, Rec, AD)$

- ▶ Normal times: $\mathcal{W}(\text{policy}, 0, 0) = 1$ (for $\Delta \mathbf{c}_{it} \approx 0$)

Welfare results

	Stimulus check	UI extension	Tax cut
$\mathcal{W}(\text{policy}, Rec = 0, AD = 0)$	0.96	0.85	0.99
$\mathcal{W}(\text{policy}, Rec = 1, AD = 0)$	1.00	1.83	0.97
$\mathcal{W}(\text{policy}, Rec = 1, AD = 1)$	1.35	2.15	1.11

- Normal times: Welfare of UI extension < 1 due to concavity of $u(\cdot)$

Relatively large change in $\mathbf{c}_{it}$ for small number of households

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- ▶ $AD = 1$: Stimulating spending during recession increases measure for all policies

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- ▶ The tax cut is both poorly targeted and may yield substantial spending after the recession is over
- ▶ Framework can be used to evaluate other candidate policies

Thank you for your attention!

- ▶ Access the paper, presentation slides and code at:
<https://github.com/econ-ark/HAFiscal>



Appendix

Parameters — same for all types

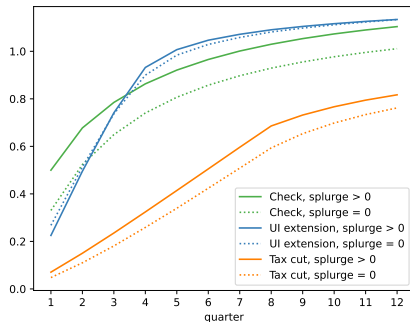
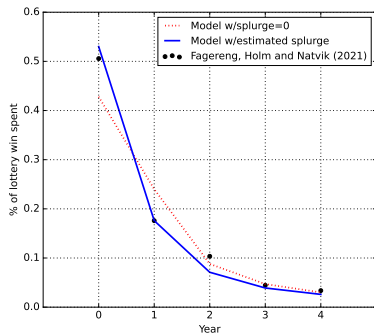
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Standard deviation of transitory shock	σ_{ξ}	0.346
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Unemployment benefits replacement rate (share of PI)	ρ_b	0.7
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Avg. duration of unemp. benefits in normal times (quarters)		2
Avg. duration of unemp. spell in normal times (quarters)		1.5
Consumption elasticity of aggregate demand effect	κ	0.3

Go back

Parameters describing the policies

Parameters describing policy experiments	
Parameter	Value
Change in unemployment rates in a recession	$\times 2$
Expected unemployment spell in a recession	4 quarters
Average length of recession	6 quarters
Size of stimulus check	\$1,200
PI threshold for reducing check size	\$100,000
PI threshold for not receiving check	\$150,000
Extended unemployment benefits	4 quarters
Length of payroll tax cut	8 quarters
Income increase from payroll tax cut	2 percent
Belief (probability) that tax cut is extended	50 percent

Robustness: Model w/o splurge consumption



	Stimulus check	UI extension	Tax cut
$\mathcal{W}(\text{policy}, Rec = 1, AD = 1)$	1.27(1.35)	2.12(2.15)	1.09(1.11)